

KULLENS FYR, Sweden

WMO: 026060

Lat: 56.3007N

Lon: 12.4520E

Elev: 72

StdP: 100.46

Time Zone: 1.00 (EUC)

Period: 86-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-8.1	-6.1	-14.0	1.1	-6.8	-11.1	1.5	-4.3	24.5	5.3	22.8	6.0	5.3	20	0.840

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.3	23.7	16.9	22.0	16.4	20.6	15.7	18.3	21.8	17.3	20.6	16.5	19.6	4.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.0	12.2	20.1	15.9	11.4	19.1	15.0	10.7	18.3	52.1	22.0	48.9	20.6	46.3	19.6	21.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.0	18.8	15.9	DB	-7.3	26.4	4.8	2.3	-10.8	28.0	-13.6	29.4	-16.3	30.7	-19.8	32.3	(4)
(5)			WB	-8.5	19.5	4.5	1.3	-11.8	20.4	-14.4	21.2	-16.9	21.9	-20.2	22.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	1.2	0.7	2.4	6.1	11.3	14.4	17.0	16.2	12.8	9.0	4.9	2.4	(6)	
	DBStd	6.45	3.72	3.51	3.02	3.45	3.13	2.77	2.58	2.13	1.99	2.69	2.50	2.72	(7)	
	HDD10.0	1359	273	259	235	130	18	1	0	0	3	49	154	236	(8)	
	HDD18.3	3712	532	492	493	368	218	121	59	75	164	289	404	494	(9)	
	CDD10.0	718	0	0	0	12	59	134	216	191	88	18	0	0	(10)	
	CDD18.3	29	0	0	0	0	1	3	17	8	0	0	0	0	(11)	
	CDH23.3	71	0	0	0	0	4	5	49	12	0	0	0	0	(12)	
	CDH26.7	7	0	0	0	0	0	0	6	1	0	0	0	0	(13)	
(14) Wind	WSAvg	7.8	9.8	9.0	8.7	6.5	6.1	6.3	6.6	7.3	7.9	8.2	8.3	9.1	(14)	
(15) Precipitation	PrecAvg	658	52	42	35	39	46	69	64	77	66	62	54	55	(15)	
	PrecMax	861	97	87	80	70	91	144	130	184	140	117	101	105	(16)	
	PrecMin	469	4	4	8	9	17	6	5	27	13	34	16	20	(17)	
	PrecStd	96	26	19	20	18	22	35	39	41	36	23	20	23	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.9	8.7	10.2	21.3	23.1	24.0	27.3	24.9	19.6	15.6	9.9	8.6	(19)	
		MCWB	5.9	7.2	7.2	12.7	16.0	15.8	19.0	17.2	15.7	13.1	8.8	7.8	(20)	
	2%	DB	6.1	6.7	7.9	17.0	20.1	21.8	24.6	22.5	17.4	14.1	9.2	7.3	(21)	
		MCWB	5.3	5.5	5.5	10.2	13.2	16.0	17.2	17.0	14.0	12.3	8.2	6.5	(22)	
	5%	DB	5.6	5.6	6.7	13.8	18.4	20.1	22.6	20.8	16.2	13.1	8.6	6.5	(23)	
		MCWB	4.7	4.4	5.1	9.0	12.2	15.4	16.8	16.7	13.5	11.5	7.6	5.7	(24)	
	10%	DB	5.1	4.8	5.8	10.7	16.5	18.7	21.0	19.3	15.4	12.3	8.0	5.8	(25)	
		MCWB	4.2	3.7	4.5	7.7	11.3	14.9	16.2	15.7	12.9	10.8	6.8	5.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	7.5	7.4	12.7	16.3	17.8	20.3	19.5	16.2	14.1	9.2	7.9	(27)	
		MCDB	6.5	8.4	9.8	20.7	22.3	20.8	24.7	23.1	18.8	15.2	9.5	8.4	(28)	
	2%	WB	5.5	5.6	6.2	10.6	13.9	17.0	18.7	18.1	15.0	12.6	8.5	6.7	(29)	
		MCDB	5.9	6.4	7.4	15.5	18.5	19.8	22.5	21.1	16.5	13.6	9.0	7.1	(30)	
	5%	WB	4.9	4.6	5.4	9.4	12.8	16.1	17.7	17.0	14.1	12.0	7.7	5.9	(31)	
		MCDB	5.4	5.4	6.4	13.4	17.1	19.2	21.3	19.9	15.7	12.9	8.3	6.4	(32)	
	10%	WB	4.4	3.9	4.7	7.8	11.7	15.2	16.7	16.1	13.3	11.2	7.1	5.1	(33)	
		MCDB	5.0	4.7	5.7	10.5	15.6	18.4	20.1	18.9	15.0	12.2	7.8	5.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	2.4	2.4	3.0	4.2	4.8	4.3	4.3	3.7	3.1	3.0	2.4	2.4	(35)	
		MCDBR	2.4	2.6	4.0	7.8	7.9	6.7	6.8	5.3	4.2	3.7	2.6	2.5	(36)	
	5% WB	MCWBR	2.6	2.4	3.0	4.4	4.9	4.3	4.1	3.5	3.0	3.1	2.7	2.6	(37)	
		MCWBR	2.3	2.5	3.7	7.0	6.8	5.4	5.5	4.7	3.9	3.3	2.5	2.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.308	0.331	0.351	0.385	0.382	0.374	0.397	0.397	0.367	0.350	0.330	0.297	(40)	
		taud	2.283	2.317	2.337	2.302	2.378	2.425	2.377	2.382	2.446	2.411	2.326	2.253	(41)	
	Ebn at Noon		586	726	807	831	858	869	838	811	787	697	552	494	(42)	
		Edh at Noon	53	75	93	112	110	107	110	103	84	68	51	42	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.42	1.01	2.37	4.00	5.21	5.50	5.10	4.10	2.88	1.39	0.52	0.28	(44)	
	RadStd		0.08	0.16	0.24	0.53	0.50	0.46	0.61	0.34	0.32	0.21	0.10	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (68 neighbors)		+0.42	N/A	N/A	N/A	N/A	N/A	-115	-169	N/A	N/A

Nomenclature: See separate page